

Assignment # 02

Title : CSC103- PROGRAMMING FUNDAMENTALS



Submitted to: MR KHURRAM IQBAL

Submitted by: ZOHA KHALID

Registration no.: CIIT/FA25-BDS-054/ISB

Class/Section: BDS/1A

Submission Date: October 16, 2025

Department of COMPUTER SCIENCE

COMSATS University Islamabad

MAIN MENU:

```
import java.util.Scanner;

import java.util.Random;

public class Assignment2 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int choice;

        do {

            System.out.println("\n===== MAIN MENU (Assignment-2) =====");

            System.out.println("1 - Question 1");

            System.out.println("2 - Question 2 ");

            System.out.println("3 - Question 3 ");

            System.out.println("4 - Question 4 ");

            System.out.println("5 - Question 5 ");

            System.out.println("0 - Exit");

            System.out.print("Enter your option (0-5): ");

            while (!sc.hasNextInt()) {

                System.out.print("Invalid input! Enter a number (0-5): ");

                sc.next();

            }

            choice = sc.nextInt();

            sc.nextLine(); // clear input buffer

            switch (choice) {
```

```
    case 1:
        question1(sc);
        break;
    case 2:
        question2(sc);
        break;
    case 3:
        question3();
        break;
    case 4:
        question4(sc);
        break;
    case 5:
        question5(sc);
        break;
    case 0:
        System.out.println("Exiting program. Goodbye!");
        break;
    default:
        System.out.println("Invalid choice! Please enter between 0–5.");
}
} while (choice != 0);
sc.close();
}
```

//Question 1: SNN VALIDATION

```
public static void question1(Scanner sc) {  
    String again;  
    do {  
        System.out.println("\n--- Executing Question 1 ---");  
        System.out.print("Enter a SSN (DDD-DD-DDDD): ");  
        String ssn = sc.nextLine();  
  
        if (ssn.matches("\\d{3}-\\d{2}-\\d{4}"))  
            System.out.println(ssn + " is a valid social security number.");  
        else  
            System.out.println(ssn + " is an invalid social security number.");  
        System.out.print("Press 'y' to try again or 'b' to go back: ");  
        again = sc.nextLine();  
    } while (again.equalsIgnoreCase("y"));  
}
```

```
--- Main Menu (Assignment-2) ---  
1 - Question 1  
2 - Question 2  
3 - Question 3  
4 - Question 4  
5 - Question 5  
0 - Exit  
Enter your option: 1  
  
--- Executing Question 1 ---  
Enter a SSN (DDD-DD-DDDD): 234 67 458  
234 67 458 is an invalid social security number.  
Press 'y' to try again or 'b' to go back: b
```

//Question 2: Major and Status

```
public static void question2(Scanner sc) {  
    String again;  
    do {  
        System.out.println("\n--- Executing Question 2 ---");  
        System.out.print("Enter two characters (Major + Year): ");  
        String input = sc.nextLine().toUpperCase();  
  
        if (input.length() != 2) {  
            System.out.println("Invalid input length!");  
        } else {  
            char major = input.charAt(0);  
            char year = input.charAt(1);  
            String majorName = "";  
            String yearName = "";  
  
            if (major == 'M') majorName = "Mathematics";  
            else if (major == 'C') majorName = "Computer Science";  
            else if (major == 'I') majorName = "Information Technology";  
  
            if (year == '1') yearName = "Freshman";  
            else if (year == '2') yearName = "Sophomore";  
            else if (year == '3') yearName = "Junior";  
            else if (year == '4') yearName = "Senior";  
  
            if (majorName.equals("") || yearName.equals(""))
```

```
        System.out.println("Invalid input!");  
    else  
        System.out.println(majorName + " " + yearName);  
    }  
    System.out.print("Press 'y' to try again or 'b' to go back: ");  
    again = sc.nextLine();  
} while (again.equalsIgnoreCase("y"));  
}
```

```
--- Main Menu (Assignment-2) ---  
1 - Question 1  
2 - Question 2  
3 - Question 3  
4 - Question 4  
5 - Question 5  
0 - Exit  
Enter your option: 2  
  
--- Executing Question 2 ---  
Enter two characters (Major + Year): M1  
Mathematics Freshman  
Press 'y' to try again or 'b' to go back: y  
  
--- Executing Question 2 ---  
Enter two characters (Major + Year): S3  
Invalid input!
```

//Question 3: Vehicle Plate Generator:

```
public static void question3() {  
    System.out.println("\n--- Executing Question 3 ---");  
    Random rand = new Random();  
    String letters = "";  
    String digits = "";  
  
    for (int i = 0; i < 3; i++)  
        letters += (char) ('A' + rand.nextInt(26));  
    for (int i = 0; i < 4; i++)  
        digits += rand.nextInt(10);  
    System.out.println("Generated plate number: " + letters + digits);  
}  
    System.out.print("Press 'y' to try again or 'b' to go back: ");  
    again = sc.nextLine();  
} while (again.equalsIgnoreCase("y"));  
}
```

```
--- Main Menu (Assignment-2) ---  
1 - Question 1  
2 - Question 2  
3 - Question 3  
4 - Question 4  
5 - Question 5  
0 - Exit  
Enter your option: 3  
  
--- Executing Question 3 ---  
Generated Vehicle Plate: YFX0194
```

//Question 4: Decimal to Binary

```
public static void question4(Scanner sc) {  
    System.out.println("\n--- Executing Question 4 ---");  
    System.out.print("Enter a decimal integer: ");  
    int number = sc.nextInt();  
    sc.nextLine();  
    String binary = "";  
    int n = number;  
    if (n == 0)  
        binary = "0";  
    else {  
        while (n > 0) {  
            binary = (n % 2) + binary;  
            n = n / 2; }  
        System.out.println("Binary value: " + binary); }  
    System.out.print("Press 'y' to try again or 'b' to go back: ");  
    again = sc.nextLine();  
    } while  
(again.equalsIgnoreCase("y")); }
```

```
--- Main Menu (Assignment-2) ---  
1 - Question 1  
2 - Question 2  
3 - Question 3  
4 - Question 4  
5 - Question 5  
0 - Exit  
Enter your option: 4  
  
--- Executing Question 4 ---  
Enter a decimal number: 9  
Binary value: 1001
```


//Question 5: Global Warming Quiz

```
public static void question5(Scanner sc) {  
    String again;  
    do {  
        System.out.println("\n--- Executing Question 5 (Global Warming Quiz) ---");  
  
        String[] questions = {  
            "1. Main gas responsible for global warming?",  
            "2. Who shared the 2007 Nobel Peace Prize with Al Gore?",  
            "3. What do global warming skeptics claim?",  
            "4. Goal of the Paris Agreement?",  
            "5. Which renewable energy reduces greenhouse gases?"  
        };  
        String[] options = {  
            {"1. Oxygen", "2. Nitrogen", "3. Carbon Dioxide", "4. Helium"},  
            {"1. NASA", "2. WHO", "3. IPCC", "4. UNESCO"},  
            {"1. It's a natural cycle", "2. It's fictional", "3. Only humans cause it", "4. Earth is  
cooling"},  
            {"1. Reduce poverty", "2. Limit temperature rise", "3. Stop deforestation", "4. Build  
dams"},  
            {"1. Coal", "2. Oil", "3. Solar", "4. Diesel"}  
        };  
  
        int[] correctAnswers = {3, 3, 1, 2, 3};  
        int score = 0;
```

```
for (int i = 0; i < questions.length; i++) {  
    System.out.println("\n" + questions[i]);  
    for (String opt : options[i]) {  
        System.out.println(opt);  
    }  
    System.out.print("Enter your answer (1-4): ");  
    while (!sc.hasNextInt()) {  
        System.out.print("Invalid! Enter 1-4: ");  
        sc.next();  
    }  
    int ans = sc.nextInt();  
    sc.nextLine();  
  
    if (ans == correctAnswers[i])  
        score++;  
}  
System.out.println("\nYou got " + score + " correct answers.");  
if (score == 5)  
    System.out.println("Excellent!");  
else if (score == 4)  
    System.out.println("Very Good!");  
else {  
    System.out.println("Time to brush up on your knowledge of global warming.");  
    System.out.println("Visit: www.climate.nasa.gov | www.ipcc.ch");  
}  
System.out.print("Press 'y' to try again or 'b' to go back: ");
```

```
        again = sc.nextLine();

    } while (again.equalsIgnoreCase("y"));

}

}
```

```
--- Main Menu (Assignment-2) ---
1 - Question 1
2 - Question 2
3 - Question 3
4 - Question 4
5 - Question 5
0 - Exit
Enter your option: 5

--- Executing Question 5 (Global Warming Quiz) ---

1. Main gas responsible for global warming?
1. Oxygen
2. Nitrogen
3. Carbon Dioxide
4. Helium
Enter your answer (1?4): 3

2. Who shared the 2007 Nobel Peace Prize with Al Gore?
1. NASA
2. WHO
3. IPCC
4. UNESCO
Enter your answer (1?4): 3

3. What do global warming skeptics claim?
1. It?s a natural cycle
2. It?s fictional
3. Only humans cause it
4. Earth is cooling
Enter your answer (1?4): 1

4. Goal of the Paris Agreement?
1. Reduce poverty
2. Limit temperature rise
3. Stop deforestation
4. Build dams
Enter your answer (1?4): 2

5. Which renewable energy reduces greenhouse gases?
1. Coal
2. Oil
3. Solar
4. Diesel
Enter your answer (1?4): 3

You got 5 correct answers.
Excellent!
```